

3.2 Haemoglobin Spectrophotometry and Electrophoresis

▼ What is the purpose of electrophoresis?

- To **separate molecules on the basis of their charge or molecular weight**

▼ How many RBCs are typically in one litre of blood?

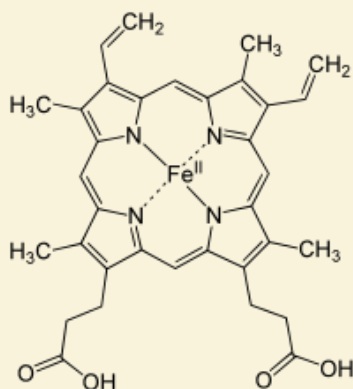
- **3.5×10^{12}**
 - Each RBC has about **640 million haemoglobin molecules**

▼ What is the function of a haem group in haemoglobin?

- To carry oxygen.

▼ What does a haem group in haemoglobin contain?

- **Fe²⁺** surrounded by a **porphyrin ring**.



▼ How does haemoglobin aid in the transportation of oxygen?

- The **Fe²⁺** of the haem group can **bind to oxygen** and form **oxyhaemoglobin**
- The O₂ is then transported from the lung to whichever tissue requires it
- It detaches from the Fe²⁺ and deoxyhaemoglobin is reformed.

▼ Describe how the protein (globin) part of a haemoglobin molecule is structured.

- It usually forms using **four** of eight possible **functional globin chains**
- These chains are arranged in "clusters" based on the genes involved in the coding.
- These are alpha clusters and beta clusters

▼ Which globin chains are used to form haemoglobin A (normal adult haemoglobin)?

- 2 alpha and 2 beta subunits

• **Hb A $\alpha_2\beta_2$ 96 - 98%**

▼ What is haemoglobin A2?

- A form of adult haemoglobin that is typically found in higher amounts in sufferers of beta-thalassaemia
 - All of us have haemoglobin A2, just in lower amounts

▼ Which globin chains are used to form haemoglobin A2 (adult haemoglobin)?

- 2 alpha and 2 delta chains

Hb A₂ $\alpha_2\delta_2$ 1.5 - 3.2%

▼ Which globin chains are used to form haemoglobin F (foetal haemoglobin)?

- 2 alpha and 2 gamma subunits.
 - "gamma" sounds like "mama" which is what babies say

Hb F $\alpha_2\gamma_2$ 0.5 - 0.8%

▼ What is myoglobin?

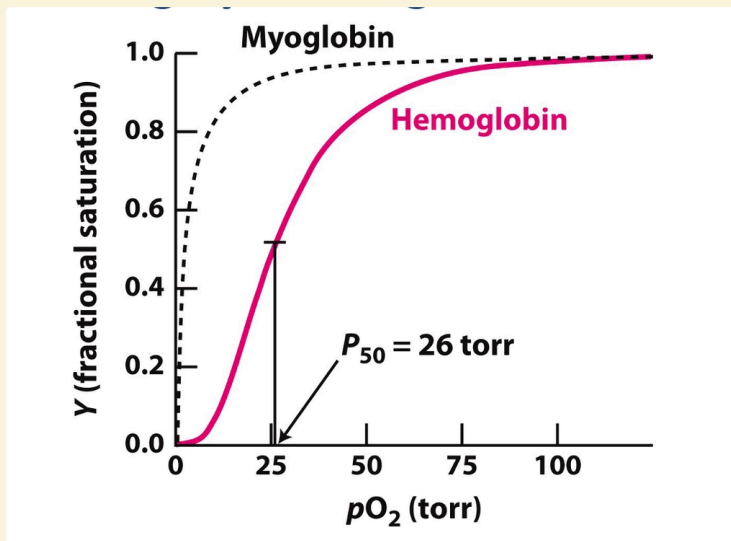
- An oxygen-binding protein found in muscle cells

▼ When referring to oxygen and haemoglobin binding to each other, what does P50 mean?

- The **partial pressure value** at which **50% of haem groups are able to be bound to oxygen.**
 - Partial pressure is often measured in torr

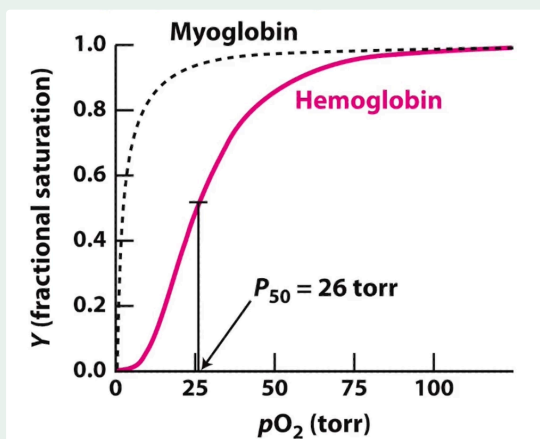
▼ Which of myoglobin and haemoglobin have a higher affinity for oxygen and how can we use P₅₀ to determine this?

- **Myoglobin** (the sigmoid curve has shift to the left)
- The **P₅₀ value of myoglobin is much lower than than of haemoglobin**. This means a **higher proportion of myoglobin binds to oxygen at lower partial pressures**.
 - However, it also means myoglobin is poor at releasing oxygen under these lower partial pressures



▼ Describe the mechanism through which haemoglobin is able to bind with oxygen and release it relevantly.

- Pressure is very high (~ 100 torr) in the lungs, and at these levels, oxygen is very easily saturated (bound) by haemoglobin where oxygen is collected from
- When it reaches the tissues around the body, the pressure is much lower and so a lot of this oxygen is released to where it is required, **but not all of it is**



▼ Describe what is meant by “positive cooperativity” when it comes to oxygen binding by haemoglobin.

- When **one haem group is able to bind onto oxygen**, this **makes it easier for the other haem groups to also bind oxygen**, until all the haem groups are bound/saturated.

▼ What is 2,3-BPG (2,3-bisphosphoglycerate)?

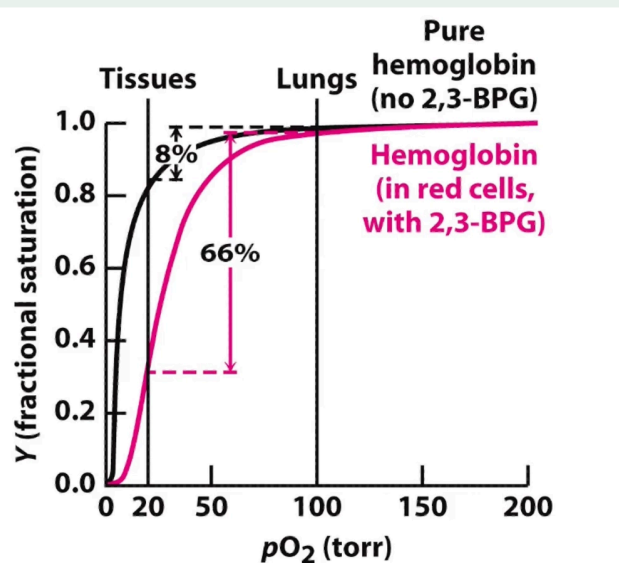
- It is an **allosteric effector (allo - other, steros - structure)** which **binds to deoxyhaemoglobin** at a site **distant from the haem groups**
- It is found in red blood cells and is used to **stabilise the structure** and **reduce the haemoglobin’s oxygen affinity** (at lower pressures)

▼ What is an allosteric effector?

- A component which can bind to another component to alter the latter’s structure **whilst having a functional impact on the latter component**.

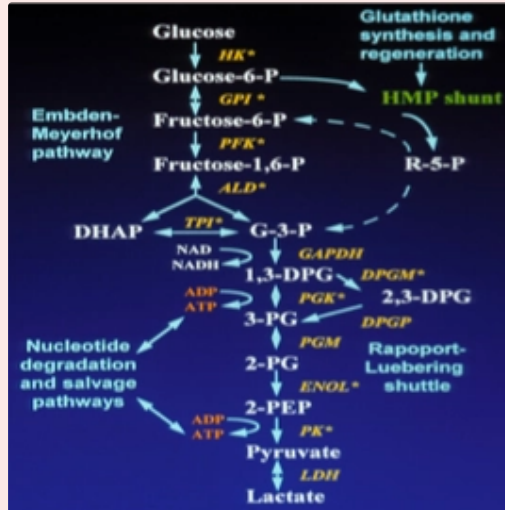
▼ Describe the importance of 2,3-BPG (2,3-bisphosphoglycerate).

- It is important because without it, **oxygen affinity at the lower pressures found in the tissues would be too high**, and so **oxygen detaching from the structure would be a difficult process**.



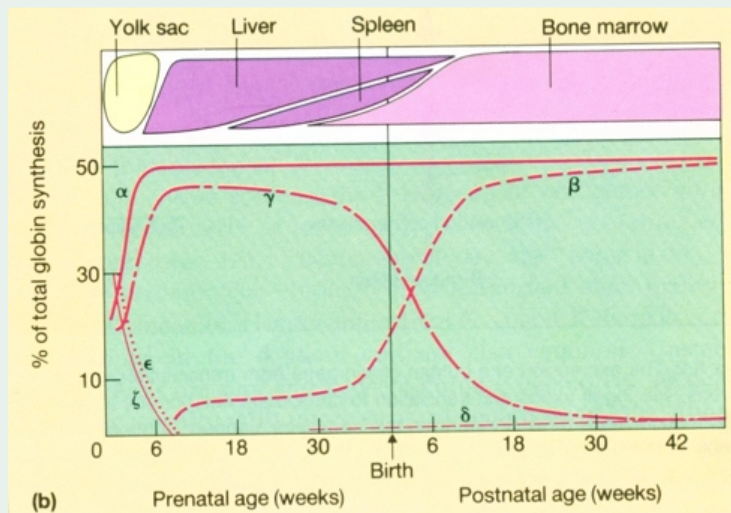
▼ How is 2,3-BPG generated?

- By the **Rapoport-Luebering shuttle**.
 - It is generated from **1,3-bisphosphoglycerate** seen in step 6 of glycolysis



▼ What is the change in gene expression that occurs to allow the production of haemoglobin A after birth in fetuses?

- **Instead of alpha and gamma chains** being the predominantly produced protein chains, **alpha and beta chains become the most predominantly produced**

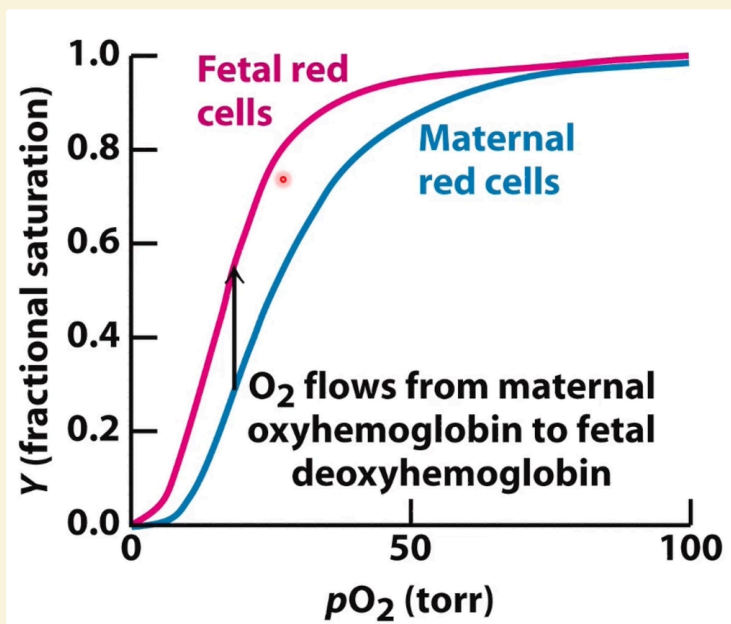


▼ Why is haemoglobin F produced in fetuses?

- To help **facilitate the transfer of oxygen** from the maternal RBCs to the foetal RBCs.

▼ How does haemoglobin F help with this maternal-foetal oxygen exchange?

- It has a **gamma chain** which has an **amino acid alteration** compared to the beta chain present in haemoglobin A
- This means that it has a **lower affinity for 2,3-BPG** than haemoglobin A
 - Since **2,3-BPG** is an allosteric effector **that decreases haemoglobin's affinity for oxygen**, a lower affinity from HbF for 2,3-BPG means it has a **higher O₂ affinity**
- This aids with the **uptake of oxygen from maternal RBCs**.

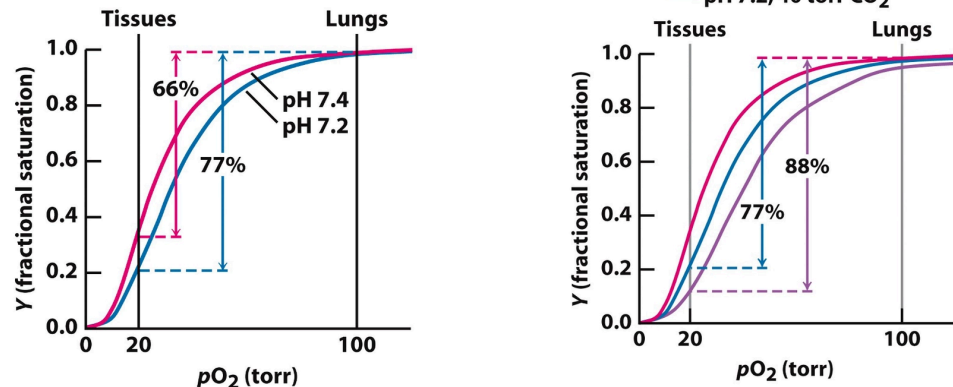


▼ What is the Bohr effect?

- It describes how **H⁺** and **CO₂** (products of lactate regeneration and pyruvate decarboxylation) can act as **allosteric effectors** by **binding to oxyhaemoglobin**
- They do this by **decreasing the affinity of oxyhaemoglobin for oxygen** and aiding greater release into tissues at lower pressures

- ▼ Explain the different Bohr effects using the graph below.

The Bohr effect – H^+ and CO_2 facilitate O_2 release from oxyhaemoglobin



► Answer

- ▼ Describe the mutation that causes the erythrocyte abnormality seen in sickle cell anaemia.

- **Glutamate** is altered to **valine at position 6** in the haemoglobin polypeptide sequence.
- Glutamate is a negatively charged (hydrophilic) amino acid but valine is neutral
- This means that the haemoglobin chain in non-sufferers is more negative than that in sickle cell anaemia sufferers

- ▼ Describe how electrophoresis can help us to discover this abnormality.

- Negatively charged specimen (i.e normal HbA) will move to the positive electrode in electrophoresis and vice-versa
- Once we notice that a sample of haemoglobin does not move to the positive electrode as much as it should (i.e HbS which stays in middle), we can begin to suspect an alteration in the amino acid sequence which has affected the polarity of the protein.

- ▼ In electrophoresis of haemoglobin samples, what is the medium we used?

- Cellulose acetate.

- ▼ What is the pH of the buffer solution and why?

- 9.5.
- We use this pH because the haemoglobin has an overall negative charge in these conditions and so migrates to the anode.

▼ Which surface of the glossy and matt filter paper should you apply solution on in gel electrophoresis?

- **Matt surface**

▼ What was the purpose of our first spectrophotometry experiment where we measured the absorbances of oxyhaemoglobin at different wavelengths?

- To **identify the wavelength which gives us the highest absorbance and hence would be best to use when analysing oxyhaemoglobin concentrations for the next experiment**

▼ Explain what we did in our second experiment to determine the oxyhaemoglobin concentrations of two unknown solutions.

- We used a **variation of known concentrations of haemoglobin solution** and measured their absorbances
 - i.e 0.2mg/ml, 0.4mg/ml, 0.6mg/ml, 0.8mg/ml, 1.0mg/ml
- This was done to **create a standard curve of these known oxyhaemoglobin concentrations (x-axis) against absorbance (y-axis)**
- We then got the **two solutions of unknown oxyhaemoglobin concentration, measured their absorbances and using the standard curve we could determine their oxyhaemoglobin concentrations**

▼ Explain how spectrophotometry can be used in the study of proteins.

- Studying the **absorbance of different proteins can help us to understand their concentrations (using the Beer-Lambert Law)**
- Understanding the concentrations of substances in non-suffering individuals and comparing them to those in sufferers, can help us to understand abnormalities in different clinical contexts.

▼ What is the Beer-Lambert Law?

- It explains how the **absorbance of a solution is directly proportional to the concentration of the absorbing material within it and to the distance (or path length) travelled by the light through the sample.**
- It describes the equation $A = E \times c \times \ell$
- **A** is the **absorbance of the solution**
- **c** is the **concentration of the absorbing substance** (typically in mol/L)
- **ℓ** is the **path length (in cm)**
- **E** is the extinction coefficient, a constant for the substance being measured at that particular wavelength.

▼ What is a pulse oximeter used for?

- **To measure oxygen saturation** (what proportion of haemoglobin is bound to oxygen)
- **Normal oxygen saturation is between 96% and 99%**
 - If there is **low oxygen saturation (<94%)**, this means **more haemoglobin is unbound to oxygen** and this could be indicative of **hypoxaemia (low blood oxygen)**

▼ Why are low levels of carbon monoxide so dangerous to the body?

- **Haemoglobin has a 200-fold greater affinity for carbon monoxide** than oxygen
- This means it can readily outcompete the oxygen for binding to the haem groups of haemoglobin
- This can lead to insufficient oxygen being delivered to cells and tissues around the body, and hence death
 - Levels of 0.2% carbon monoxide can lead to death within an hour or two.

▼ What is methaemoglobin (MetHb)?

- This is a form of haemoglobin that **contains the oxidised, ferric form of iron (Fe³⁺) instead of the normal ferrous Fe²⁺**
 - It is generated when this **Fe²⁺ is oxidised to Fe³⁺**

▼ What are the clinical consequences of having methaemoglobin and why?

- **Fe³⁺ binds to oxygen more strongly** than Fe²⁺
- This means that **methaemoglobin's (Fe³⁺) affinity for oxygen is much higher**, even at lower partial pressures
- This is **impaired oxygen binding**, and can lead to **not enough oxygen**